

A whole lot of reductions

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1 Preliminaries

Definition 1. A **tour** in a graph $G = (V, E)$ is a sequence of vertices and edges adjacent to them in the graph that spans every vertex. We usually refer to the multiset of edges defined by a tour as T . The **cost** of a tour T is the sum of the weights of the edges in T , denoted as $c(T) = \sum_{e \in T} c(e)$.

This definition is equivalent to saying that a tour is a Eulerian multigraph that spans every vertex of G .

Definition 2 (Metric TSP). Let $G = (V, E)$ be a connected undirected graph. The **Metric Traveling Salesperson Problem** (Metric TSP) is defined as follows:

- **Instance:** A complete graph $G = (V, E)$ which is connected by definition, and a cost function $c : E \rightarrow \mathbb{R}_{\geq 0}$ that satisfies the triangle inequality:

$$c(u, v) \leq c(u, w) + c(w, v) \quad \text{for all } u, v, w \in V. \quad (1)$$

- **Objective:** Find a closed tour C in G such that the total weight $c(C) = \sum_{e \in C} c(e)$ is minimized.

Definition 3 (Quasi-Tour). A **quasi-tour** T_0 in a graph $G = (V, E)$ is a subset of edges in T removing edges from a finite set of closed tours. Each connected component of this multigraph is either a Eulerian multigraph or an isolated vertex.

A given multiset of edges T_0 is a quasi-tour if and only if it is Eulerian.

2 Starting point

For the reduction, we will start from a common hard to approximate problem:

Definition 4 (MAX-E3-LIN-2). The **MAX-E3-LIN-2** problem is defined as follows:

- **Instance:** A set of n boolean variables $x_1, x_2, \dots, x_n$ and a collection of m linear equations over $\mathbb{F}_2$, where each equation involves exactly three variables. Each equation is of the form:

$$x_{i_1} + x_{i_2} + x_{i_3} = b_j, \quad (2)$$

where $b_j \in \{0, 1\}$ and i_1, i_2, i_3 are distinct indices from $\{1, 2, \dots, n\}$.

- **Objective:** Find an assignment to the variables that maximizes the number of satisfied equations.

Theorem 1. For every $\varepsilon \in (0, \frac{1}{4})$ and a sufficiently large integer $k \geq k(\varepsilon)$, the following partial decision subproblem $Q(\varepsilon, k)$ of MAX-E3-LIN-2 is NP-hard: given an instance of MAX-E3-LIN-2 with m equations and exactly k occurrences of each variable, to decide if at least $(1 + \varepsilon)m$ equations can be satisfied or if at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.

To prove some inapproximability results, it is more convenient for reductions if all equations have the same right hand side. The results obtained in such case can be enforced by flipping some occurrences of variables ($\tilde{x} = 1 - x$). With that idea in mind, we can define a subproblem of $Q(\varepsilon, k)$.

Definition 5. For $b \in \{0, 1\}$, we define $Q_b(\varepsilon, k)$ as the following partial decision problem:

- **Instance:** An instance of MAX-E3-LIN-2 with m equations of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$, each variable occurring exactly k times negated and k times unnegated.
- **Objective:** Decide if at least $(1 + \varepsilon)m$ or at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.

To continue with the reduction, we will extend the NP-hardness results to a low occurrence version of w -MAX-E3-LIN-2, a weighted system of linear equations over $\mathbb{Z}_2$ with either 2 or 3 variables. We call this a hybrid system of linear equations. Similarly, the task of the w -MAX-E3-LIN-2 problem is to find an assignment that maximizes the total weight of satisfied equations.

3 Reductions to w -MAX-E3-LIN-2

3.1 From $Q(\varepsilon, k)$ to w -MAX-E3-LIN-2

Let $\varepsilon \in (0, \frac{1}{4})$ and $k > 0$ be an integer such that the problem $Q(\varepsilon, k)$ is NP-hard. Let an instance I of $Q(\varepsilon, k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$. let's assume that there is an amplifier $G = (V, E)$ with edge weights $p : E \leftrightarrow \mathbb{R}^+$, with a set of contacts D with $|D| = k$.

Now, there is a gap preserving reduction from $Q(\varepsilon, k)$ to w -MAX-E3-LIN-2 with the amplifier as a parameter. The instance I of $Q(\varepsilon, k)$ is transformed into a hybrid instance J of w -MAX-E3-LIN-2 as follows:

- For each variable $x \in \nu(I)$, we create a copy of the amplifier G , denoted G_x .
 - Inside G_x , the vertices correspond to variables in J and each edge (u, v) represents the equation $u + v = 0$ with weight $p(u, v)$.
 - Contacts in G_x represent the k occurrences of the variable x in the equations of I . Each contact is associated with one occurrence of x in I .
- Every equation in I of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$ also belongs to J with weight 1.

Observe that the reduction above from I to J is a gap preserving reduction. Indeed, there is a dependency between the number of equations satisfied in I and the total weight of satisfied equations in J .

- If at least $(1 + \varepsilon)m$ equations can be satisfied in I , then there is an assignment that satisfies all the equations in the amplifiers and at least $(1 - \varepsilon)m$ equations of weight 1, leading to a total weight of satisfied equations of at least $W_1 = k \cdot p(E) + (1 - \varepsilon)m$.
- If at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied in I , then any assignment can satisfy at most $\frac{1}{2}m - \varepsilon m$ equations of weight 1, leading to a total weight of satisfied equations of at most $W_2 = k \cdot p(E) + (\frac{1}{2} - \varepsilon)m$.

If we look at these problems as minimum unsatisfiability problems, where OPT' is the minimum weight of unsatisfied equations over all assignments, then $\text{OPT}'(I) = \text{OPT}'(J)$. Any assignment to variables from $\nu(I)$ generate an assignment to variables of J naturally, as the value of $x \in \nu(I)$ will be assigned to all the variables in G_x . These are called standard. Then, $\text{OPT}'(J) \leq \text{OPT}'(I)$ because any assignment to I can be extended to a standard assignment to J with the same weight of unsatisfied equations.

To see that the optimum $\text{OPT}'(J)$ is attained on standard assignments, we use the properties of the amplifier. Any assignment φ to the variables of J can be transformed into a standard assignment φ' without increasing the weight of unsatisfied equations. For this, we assign to all variables in G_x the value as it is assigned to the majority of contact vertices in G_x by φ . The fact that G is an amplifier guarantees (by definition) that the weight of unsatisfied equations in J does not increase in this transformation. After iterating this process for all variables, the result will be a standard assignment without increasing the weight of unsatisfied equations in J . Therefore, $\text{OPT}'(J) = \text{OPT}'(I)$.

3.2 From $Q_b(\varepsilon, 2k)$ to $w\text{-MAX-E3-LIN-2}$

The previous reduction is only the starting point, and a demonstration of how amplifiers can be useful. However, for the main results in this article, it is more convenient to start from the problem $Q_b(\varepsilon, k)$, where all equations have the same right hand side. This reduction is a slight modification of the previous one. Let

$\varepsilon \in (0, \frac{1}{4})$ and $k > 0$ be an integer such that the problem $Q_b(\varepsilon, k)$ is NP-hard. Assume that $G = (V, E)$ is an amplifier with edge weights $p : E \leftrightarrow \mathbb{R}^+$ and a set of contacts D with $|D| = 2k$. Let $\{V^u, V^n\}$ be a partition of V balanced in D , so $\|D \cap V^u\| = \|D \cap V^n\| = k$. Also, let G^u and G^n be the induced subgraphs of G on V^u and V^n , respectively. Let an instance I of $Q_b(\varepsilon, 2k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$.

- For each variable $x \in \nu(I)$, we create a copy of the amplifier G , denoted G_x .
 - Any edge (u, u') in G_x^u or $G_x^{u'}$ represents the equation $u + u' = 0$ with weight $p(u, u')$. We call this cycle equations.
 - Any edge between $v \in V_x^u$ and $v' \in V_x^n$ represents the equation $v + v' = 1$ with weight $p(v, v')$. We call this matching equations.
- The contact vertices of G_x^u (and of G_x^n) represent the k occurrences of unnegated (respectively, negated) variable x in the equations of I . Every equation $x + y + z = b$ from I also belongs to J with weight 1.

This way, we produce an instance J of w -MAX-E3-LIN-2 with the same gap as in the previous reduction. As before, any assignment to variables from $\nu(I)$ generate an assignment to variables of J naturally, as the value of $x \in \nu(I)$ will be assigned to all the variables in G_x^u , and the value $1 - x$ to all the variables in G_x^n . Once again, this assignment is called standard. The previous arguments applied to the problem $Q(\varepsilon, 2k)$ can be translated to this new setting, by reversing all variable from V_x^n to their opposite.

3.3 Applying bi-wheel amplifiers

When we consider the existence of bi-wheel amplifiers $(W_{k,\tau}, p)$ and add them to the framework we have just defined, we obtain the following theorem:

Theorem 2 (Hybrid- $(W_{k,\tau}, p)$ inapproximability). *For every $\varepsilon \in (0, \frac{1}{4})$, $b \in 0, 1$ there exist instances of w -MAX-E3-LIN-2, which we call hybrid- $(W_{k,\tau}, p)$, with the following properties:*

- Each variable of the system of equations hybrid- $(W_{k,\tau}, p)$ occurs exactly 3 times.
- m equations are of the form $x + y + z = b$ with weight 1.
- $3\tau m$ equations are of the form $x + y = 0$, with weight p_c .
- $\frac{3}{2}(\tau - 1)m$ equations are of the form $x + y = 1$, with weight p_m .

It is NP-hard to decide if equations of weight at least $(1 - \varepsilon)m$ or at most $(\frac{1}{2} - \varepsilon)m$ can be satisfied by an optimal assignment.

Proof. Let $\varepsilon \in (0, \frac{1}{4})$, $b \in 0, 1$ and $k > 0$ be an integer such that the problem $Q_b(\varepsilon, 2k)$ is NP-hard. Let an instance I of $Q_b(\varepsilon, 2k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$. Then $m := |I| = \frac{2k\nu}{3}$, as each variable occurs exactly $2k$ times and each equation has 3 variables. For any fixed variable $x \in \nu(I)$, we create a copy of the bi-wheel amplifier $W_{k,\tau}$, denoted W_x . Each of these is a copy of $W_{k,\tau}$, with $2\tau k$ variables $Var(x) = \{x_j^u, x_{-j}^n\}_{j=1}^{\tau k}$. Following the definition of a biwheel amplifier, each j^u is mapped to the variable x_j^u and j^n is mapped to x_j^n . We will denote the set of cycle edges as $\mathcal{C}^u(W_x)$ and $\mathcal{C}^n(W_x)$, and the set of matching edges as $\mathcal{M}(W_x)$.

- For each matching edge (j^u, j'^n) in $\mathcal{M}(W_x)$, we create an equation $x_j^u + x_{j'}^n = 1$ with weight p_m . There are $\frac{3}{2}(\tau - 1)m$ such equations.
- For each cycle edge $(j^q, (j + 1)^q)$ in $\mathcal{C}^q(W_x)$, we create an equation $x_j^q + x_{j+1}^q = 0$ with weight p_c . We have $2k\tau\nu = 3\tau m$ such equations.

Finally, we replace the l -th unnegated appearance of x in I with the variable $x_{c^u(l)}^u := c_l^u$, and the l -th negated appearance of x in I with the variable $x_{c^n(l)}^n := c_l^n$, for each $l \in \{1, \dots, k\}$. This doesn't add equations to the system, but it just renames the variables in the equations of I to the variables in the bi-wheel amplifiers.

Then, we have produced m equations of the form $x + y + z = b$ with weight 1, $3\tau m$ equations of the form $x + y = 0$, with weight p_c , and $\frac{3}{2}(\tau - 1)m$ equations of the form $x + y = 1$, with weight p_m . The degree of the variables is 3 in any case, as each variable in the bi-wheel amplifier has degree 3, and each variable in the equations of I is replaced by a variable in the bi-wheel amplifier that also has degree 3. The gap preserving property of the reduction is a consequence of the properties of the bi-wheel amplifier, as mentioned in the previous section. $\square$